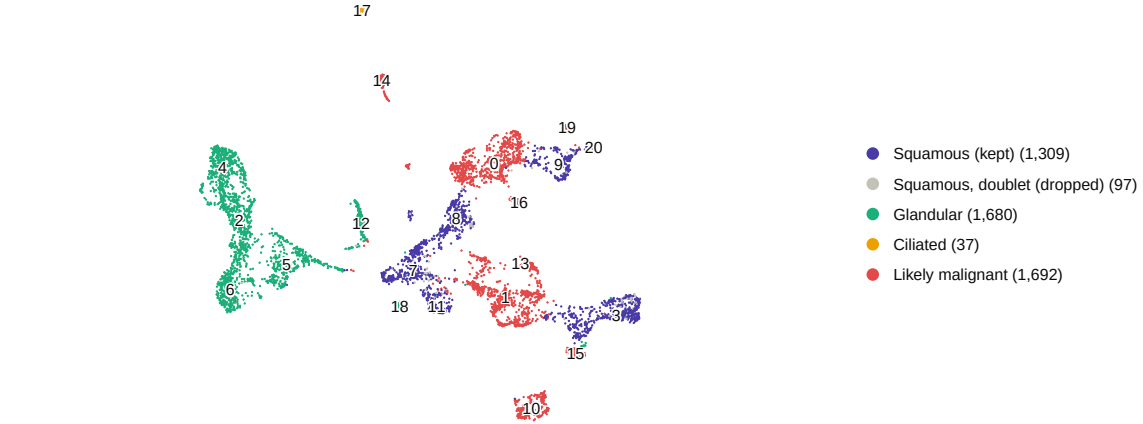
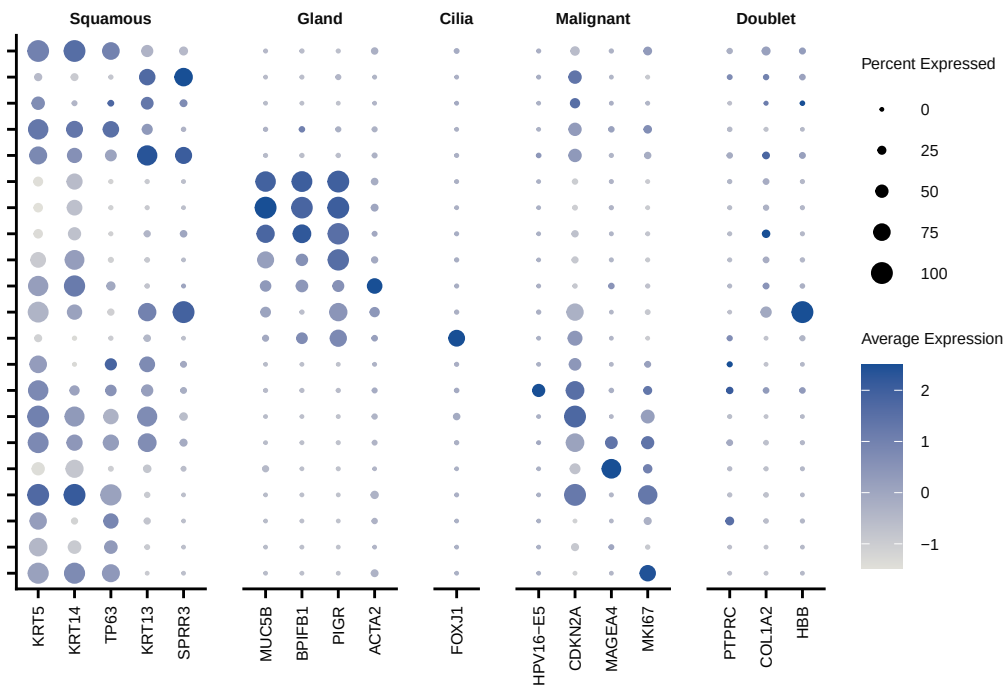
a


Atlas 'non−malignant epithelial' cells, sub−clustered
4,815 cells, 60 patients; numbers = clusters
17
14
19
20
4
0
9
16
2
12
8
13
5
7
6
1
18
11
3
15
10
Squamous (kept) (1,309)
Squamous, doublet (dropped) (97)
Glandular (1,680)
Ciliated (37)
Likely malignant (1,692)

b


Marker genes per cluster
Squamous
Gland
Cilia
Malignant
Doublet
3 Squamous
7 Squamous
8 Squamous
9 Squamous
11 Squamous
2 Glandular
4 Glandular
5 Glandular
6 Glandular
12 Glandular
18 Glandular
17 Ciliated
0 Likely malignant
1 Likely malignant
10 Likely malignant
13 Likely malignant
14 Likely malignant
15 Likely malignant
16 Likely malignant
19 Likely malignant
20 Likely malignant
KRT5
KRT14
TP63
KRT13
SPRR3
MUC5B
BPIFB1
PIGR
ACTA2
FOXJ1
HPV16−E5
CDKN2A
MAGEA4
MKI67
PTPRC
COL1A2
HBB
Percent Expressed
0
25
50
75
100
Average Expression
2
1
0
−1